

Redox Reactions

Complete chapter notes — oxidation numbers, every type of redox change, both balancing methods worked step by step, redox titrations and the electrode processes behind them.

The spine of this chapter: a redox reaction is nothing but **electron bookkeeping**. Assign oxidation numbers, see which element goes **up** (oxidation, electrons lost) and which goes **down** (reduction, electrons gained), and every question in the chapter — identify, classify, balance, titrate, predict — is answered from that one table of numbers. Oxidation and reduction always occur **together** and in exactly equal electron counts; that equality *is* the balancing method.

1 · Two Concepts of Oxidation and Reduction

	Oxidation	Reduction
Classical	Addition of oxygen or of any electronegative element; removal of hydrogen or of any electropositive element	Removal of oxygen; addition of hydrogen
Electronic	Loss of electrons	Gain of electrons
Oxidation number	Increase in oxidation number	Decrease in oxidation number

MEMORY HOOK

OIL RIG — **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain (of electrons). And remember the agents are named for what they do to the *other* substance: the **oxidising agent is itself reduced**, the **reducing agent is itself oxidised**.

The electronic definition is the general one — it covers reactions with no oxygen at all. In $2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$, sodium loses electrons (oxidised, reducing agent) and chlorine gains them (reduced, oxidising agent).

2 · Oxidation Number

Oxidation number is the charge an atom *appears* to have in a species when all the bonds are treated as fully ionic — the shared electrons of each bond assigned entirely to the more electronegative atom.

2.1 The rules, in order of priority

1. An atom in its **free element** has oxidation number **0**: Na, O₂, P₄, S₈, Cl₂.
2. A **monatomic ion** has an oxidation number equal to its charge: Na⁺ is +1, S²⁻ is -2.
3. **Fluorine is always -1** — no exceptions, it is the most electronegative element.
4. **Group 1 metals are +1**, group 2 are +2, aluminium is +3 in all their compounds.
5. **Hydrogen is +1** with non-metals, but **-1 in metal hydrides** (NaH, CaH₂, LiAlH₄).
6. **Oxygen is -2**, except: **-1 in peroxides** (H₂O₂, Na₂O₂), **- $\frac{1}{2}$ in superoxides** (KO₂), and **positive with fluorine** (+2 in OF₂, +1 in O₂F₂).
7. The **sum** of the oxidation numbers equals 0 for a neutral molecule and equals the charge for an ion.

WORKED EXAMPLE 1 — COMPUTING OXIDATION NUMBERS

(a) **Cr in Cr₂O₇²⁻**: $2x + 7(-2) = -2 \Rightarrow 2x = 12 \Rightarrow x = \mathbf{+6}$

(b) **Mn in MnO₄⁻**: $x + 4(-2) = -1 \Rightarrow x = \mathbf{+7}$

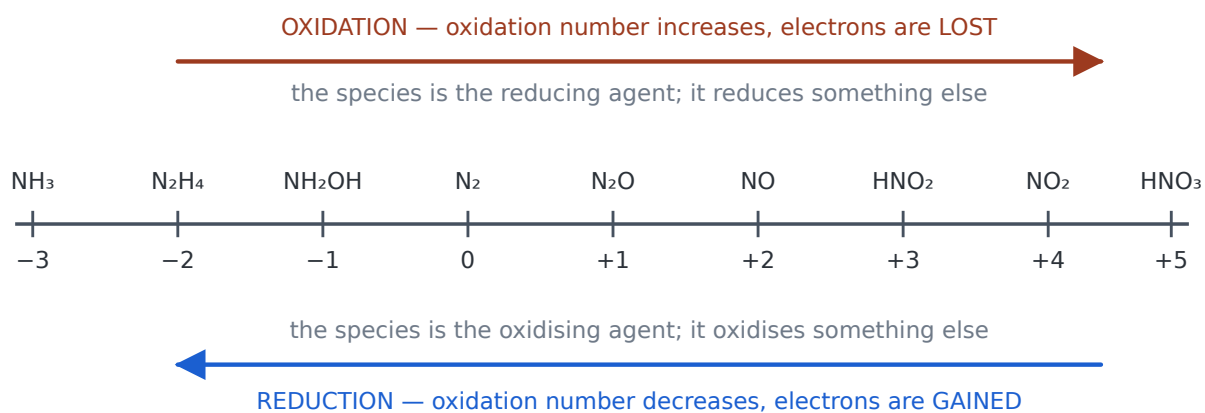
(c) **Fe in Fe₃O₄**: $3x + 4(-2) = 0 \Rightarrow x = \mathbf{+8/3}$ — a **fractional average**, because Fe₃O₄ is really FeO·Fe₂O₃, containing one Fe(II) and two Fe(III).

(d) **S in Na₂S₄O₆ (sodium tetrathionate)**: $2(+1) + 4x + 6(-2) = 0 \Rightarrow 4x = 10 \Rightarrow x = \mathbf{+2.5}$ — again an average: two of the sulfurs are at +5 and two, joined by an S-S bond, are at 0.

(e) **C in C₆H₁₂O₆**: $6x + 12(+1) + 6(-2) = 0 \Rightarrow x = \mathbf{0}$, even though the individual carbons are certainly not all zero.

OXIDATION NUMBER IS NOT VALENCY

1. Oxidation number may be **positive, negative, zero or fractional**; **valency is always a whole number and has no sign**. In CH₄ the oxidation number of carbon is -4 while its valency is 4.
2. A fractional value *always* means an **average** over atoms in genuinely different environments. Never write "the oxidation state of Fe is +8/3" without saying it is an average.
3. **Stock notation** puts the oxidation number in Roman numerals after the element: Fe(III) in Fe₂O₃, Mn(VII) in KMnO₄, Cu(I) in Cu₂O.



The oxidation-number line for nitrogen. Any species in the middle of its range can move both ways, and that is precisely the condition for **disproportionation**. An element already at its highest state (N in HNO_3) can only be reduced; at its lowest (N in NH_3) it can only be oxidised.

3 · Types of Redox Reactions

Type	Description	Example
Combination	Two species combine; redox only if an element is involved	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
Decomposition	A compound breaks into two or more products, at least one an element	$2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$ $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
Displacement (metal)	A more reactive metal displaces a less reactive one from its salt	$\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$
Displacement (non-metal)	Hydrogen or a halogen is displaced	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$
Disproportionation	The same element in a single species is both oxidised and reduced	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
Comproportionation	The reverse: two different oxidation states of one element give a single intermediate state	$5\text{Cl}^- + \text{ClO}_3^- + 6\text{H}^+ \rightarrow 3\text{Cl}_2 + 3\text{H}_2\text{O}$

3.1 Disproportionation — the type most often asked

Requirement: the element must be in an **intermediate** oxidation state, with both a higher and a lower state available to it.

Reaction	Element	Goes to
$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$	O at -1	-2 (reduced) and 0 (oxidised)
$\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$	Cl at 0	-1 and $+1$
$2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$	Cu at $+1$	$+2$ and 0
$\text{P}_4 + 3\text{OH}^- + 3\text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$	P at 0	-3 and $+1$

FREQUENTLY ASKED

Why does fluorine not disproportionate? Because fluorine is the most electronegative element and has **no positive oxidation state** — it can only be reduced from 0 to -1 , never oxidised. So F_2 with alkali gives OF_2 or O_2 (oxidising the *oxygen*), not the F^-/FO^- pair that chlorine gives.

4 · Balancing Redox Reactions

4.1 The oxidation number method

METHOD 1 Balancing by the oxidation number method

1. Write the skeletal equation and assign oxidation numbers to every atom.
2. Identify the elements whose oxidation numbers **change**, and compute the increase and the decrease **per atom**, then per formula unit.
3. Multiply the two species by whatever coefficients make the **total increase equal to the total decrease**.
4. Balance all remaining atoms except H and O by inspection.
5. Balance **oxygen** by adding H_2O , then **hydrogen** by adding H^+ (acidic medium).
6. In a **basic** medium, finish by adding as many OH^- ions to **both** sides as there are H^+ , and combine $\text{H}^+ + \text{OH}^-$ into H_2O .
7. Verify that both the **atoms** and the **total charge** balance.

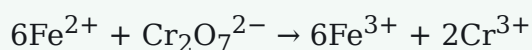
Step 3 is the whole method: electrons lost = electrons gained is the same statement as total increase = total decrease.

WORKED EXAMPLE 2 — DICHROMATE AND IRON(II) IN ACID

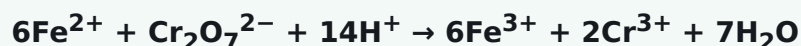
Skeleton: $\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + \text{H}^+ \rightarrow \text{Fe}^{3+} + \text{Cr}^{3+} + \text{H}_2\text{O}$

Changes: Fe goes $+2 \rightarrow +3$ — an increase of **1** per atom. Cr goes $+6 \rightarrow +3$ — a decrease of **3** per atom, and there are **2** Cr atoms, so **6** per dichromate ion.

Equalise: multiply Fe^{2+} by 6. $6 \times 1 = 1 \times 6$ ✓



Balance O with $7\text{H}_2\text{O}$ on the right, then H with 14H^+ on the left:



Charge check: left = $6(+2) + (-2) + 14(+1) = +24$; right = $6(+3) + 2(+3) = +24$.

Balanced.

4.2 The half-reaction (ion-electron) method

METHOD 2 Half-reaction method — acidic medium

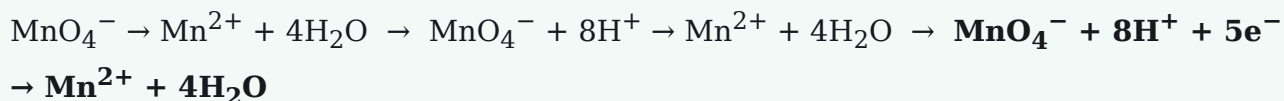
1. Split the reaction into the **oxidation half** and the **reduction half**.
2. In each half, balance all atoms **except H and O**.
3. Balance **O** by adding H_2O to the deficient side.
4. Balance **H** by adding H^+ .
5. Balance the **charge** by adding electrons — to the left of the reduction half, to the right of the oxidation half.
6. Multiply the two halves so that the **electrons cancel**, then add them together.
7. Cancel anything appearing on both sides, and check atoms and charge.

For a basic medium, do all of the above, then add OH^- to both sides equal to the number of H^+ , convert $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, and cancel duplicated water molecules.

WORKED EXAMPLE 3 — PERMANGANATE AND SULFITE IN ACID

Skeleton: $\text{MnO}_4^- + \text{SO}_3^{2-} \rightarrow \text{Mn}^{2+} + \text{SO}_4^{2-}$ (acidic)

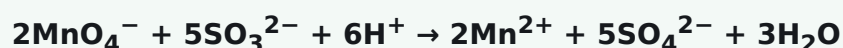
Reduction half (Mn: +7 $\rightarrow$ +2):



Oxidation half (S: +4 $\rightarrow$ +6):



Equalise electrons: reduction $\times 2$, oxidation $\times 5$ (LCM of 5 and 2 is 10), then add and cancel:

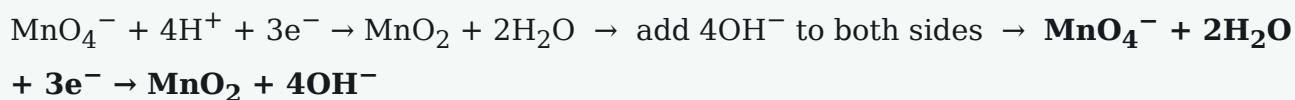


Check. O: $8 + 15 = 23$ left, $20 + 3 = 23$ right $\checkmark$. H: 6 both sides $\checkmark$. Charge: $-2 - 10 + 6 = -6$ left, $+4 - 10 = -6$ right $\checkmark$ **Balanced.**

WORKED EXAMPLE 4 — PERMANGANATE AND IODIDE IN ALKALI

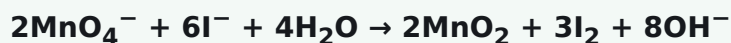
Skeleton: $\text{MnO}_4^- + \text{I}^- \rightarrow \text{MnO}_2 + \text{I}_2$ (basic)

Reduction half (Mn: +7 $\rightarrow$ +4), balanced first in acid then converted:



Oxidation half: $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$

Multiply the first by 2 and the second by 3 (six electrons each), then add:



Charge check: left = $-2 - 6 = -8$; right = -8 $\checkmark$ **Balanced.**

WHERE BALANCING GOES WRONG

1. **Never add H^+ to a basic medium** in the final equation, and never add OH^- to an acidic one.
2. **Electrons must cancel completely.** If any remain in the final equation, the halves were not multiplied correctly.
3. **Check the total charge, not only the atoms.** A charge mismatch is the fastest way to catch an error.
4. Never change a formula (writing SO_4^- for SO_4^{2-}) to make the balancing work — only coefficients may change.

5 · Redox Titrations

The end point of a redox titration is found either from a **self-indicating** reagent ($KMnO_4$, whose first permanent pink tinge marks the end point) or from an **external indicator** (starch, giving a blue-black colour with iodine).

5.1 Permanganate — a different product in each medium

Medium	Half-reaction	Electrons (n -factor)	Mn goes
Acidic (dilute H_2SO_4)	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	5	$+7 \rightarrow +2$ (colourless)
Neutral / faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$	3	$+7 \rightarrow +4$ (brown solid)
Strongly alkaline	$MnO_4^- + e^- \rightarrow MnO_4^{2-}$	1	$+7 \rightarrow +6$ (green)

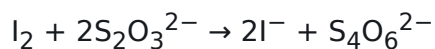
WHY SULFURIC ACID AND NEVER HCL OR HNO_3

Permanganate titrations are acidified with **dilute H_2SO_4** . Hydrochloric acid would itself be oxidised to chlorine, consuming extra $KMnO_4$ and giving a high result; nitric acid is itself an oxidising agent and would attack the reducing agent being estimated.

5.2 Dichromate and iodometry

Dichromate works only in acid, with $n = 6$: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. It is a **primary standard** — obtainable pure and stable — which $KMnO_4$ is not.

Iodometry liberates iodine, which is then titrated against standard sodium thiosulfate with starch near the end point:



Note the sulfur: +2 in thiosulfate rises to the average +2.5 in tetrathionate, so each thiosulfate loses exactly one electron.

5.3 Equivalents and n -factor

n -factor = number of electrons gained or lost per formula unit. Equivalent mass = $\frac{\text{molar mass}}{n\text{-factor}}$, Normality = Molarity $\times n$ -factor, and at the end point $N_1V_1 = N_2V_2$.

WORKED EXAMPLE 5 — A TITRATION CALCULATION

25.0 mL of an FeSO_4 solution needs 20.0 mL of 0.02 M KMnO_4 in dilute H_2SO_4 . Find the molarity of the iron(II) solution.

Reaction: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$, so the mole ratio is **1 : 5**.

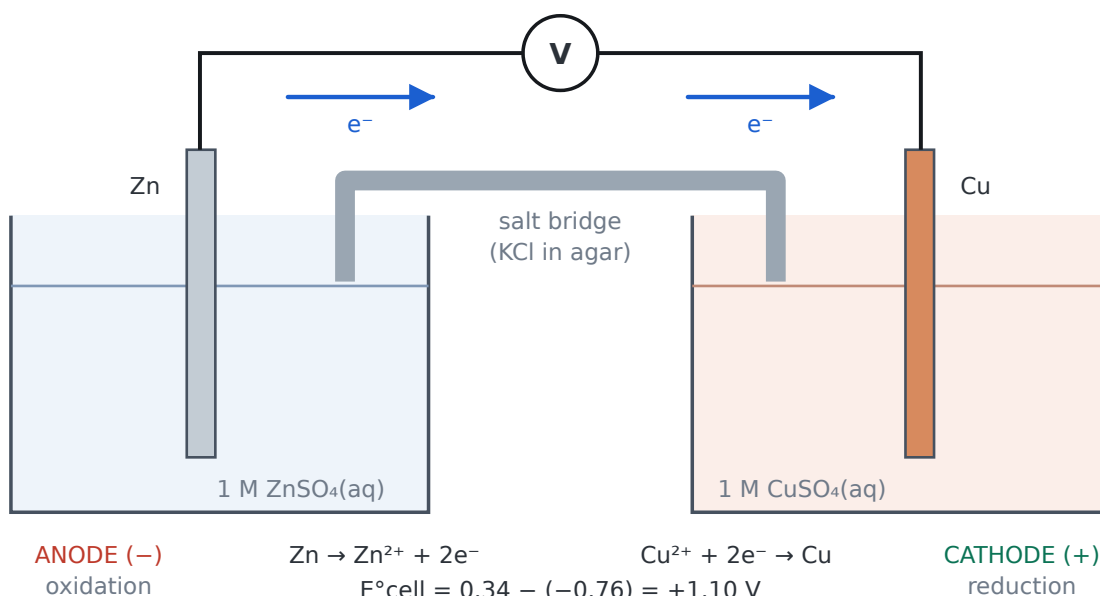
Moles of $\text{MnO}_4^- = 0.02 \times 0.0200 = 4.0 \times 10^{-4} \text{ mol} \Rightarrow \text{moles of Fe}^{2+} = 5 \times 4.0 \times 10^{-4} = 2.0 \times 10^{-3} \text{ mol}$

$$M(\text{Fe}^{2+}) = \frac{2.0 \times 10^{-3}}{0.0250} = \mathbf{0.080 \text{ M}}$$

Same answer by equivalents: $N(\text{KMnO}_4) = 0.02 \times 5 = 0.1 \text{ N}$; $N_1V_1 = N_2V_2 \Rightarrow N(\text{Fe}^{2+}) = \frac{0.1 \times 20}{25} = 0.08 \text{ N}$, and since $n = 1$ for Fe^{2+} , $M = 0.08 \text{ M}$.

6 · Redox Reactions and Electrode Processes

If the two halves of a redox reaction are physically separated, the electrons must travel through an external wire — and that current is usable work. This is a **galvanic (voltaic) cell**.


$$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$$
$$E^\circ_{\text{cell}} = 0.$$

CATHODE (+)
reduction

MEMORY HOOK

Activity series, most reactive first: **Li K Ba Ca Na Mg Al Zn Fe Ni Sn Pb (H) Cu Hg Ag Au** — "Little **K**atie **B**akes **C**akes, **N**aughty **M**aggie **A**lways **Z**aps **F**erocious **N**ice **S**nakes, **Pb**... **H**ere **C**urious **H**ungry **A**gents **A**udition." Any metal displaces every metal *below* it from solution.

7 · Where marks are lost

THE TWELVE TRAPS OF CHAPTER 7

1. **The oxidising agent is the one that gets reduced.** Write the oxidation numbers down before naming any agent.
2. **Oxidation number can be fractional; valency cannot.** A fraction always means an average over inequivalent atoms.
3. **Oxygen is not always -2** — it is -1 in peroxides, $-\frac{1}{2}$ in superoxides and **positive** in OF_2 . Fluorine, however, is always -1 .
4. **Hydrogen is -1 in metal hydrides** (NaH , CaH_2).
5. **Disproportionation needs an intermediate oxidation state.** Fluorine cannot disproportionate; nor can an element already at its highest or lowest state.
6. **Balance the charge as well as the atoms.** An equation with balanced atoms and unbalanced charge is still wrong.
7. **In basic medium, finish with OH^- , not H^+ .** Add OH^- to both sides and collapse the pairs into water.
8. **KMnO_4 gives a different product in each medium** — $n = 5$ in acid, 3 in neutral, 1 in strong alkali. Read the medium before writing the half-reaction.
9. **Acidify permanganate with dilute H_2SO_4 only**, never HCl (oxidised to Cl_2) or HNO_3 (itself an oxidant).
10. **Not every reaction with oxygen is redox, and not every redox has oxygen.** Precipitation and acid-base neutralisation involve no change in oxidation number at all.
11. **In a galvanic cell the anode is negative** (the opposite of an electrolytic cell) and oxidation always occurs there.
12. **E° values are not multiplied by stoichiometric coefficients.** Doubling a half-reaction does not double its electrode potential.

Chapter 7 of the rationalised NCERT (2023-24 onward) — the opening chapter of Chemistry Part II. Standard electrode potentials quoted at 298 K against the standard hydrogen electrode.